

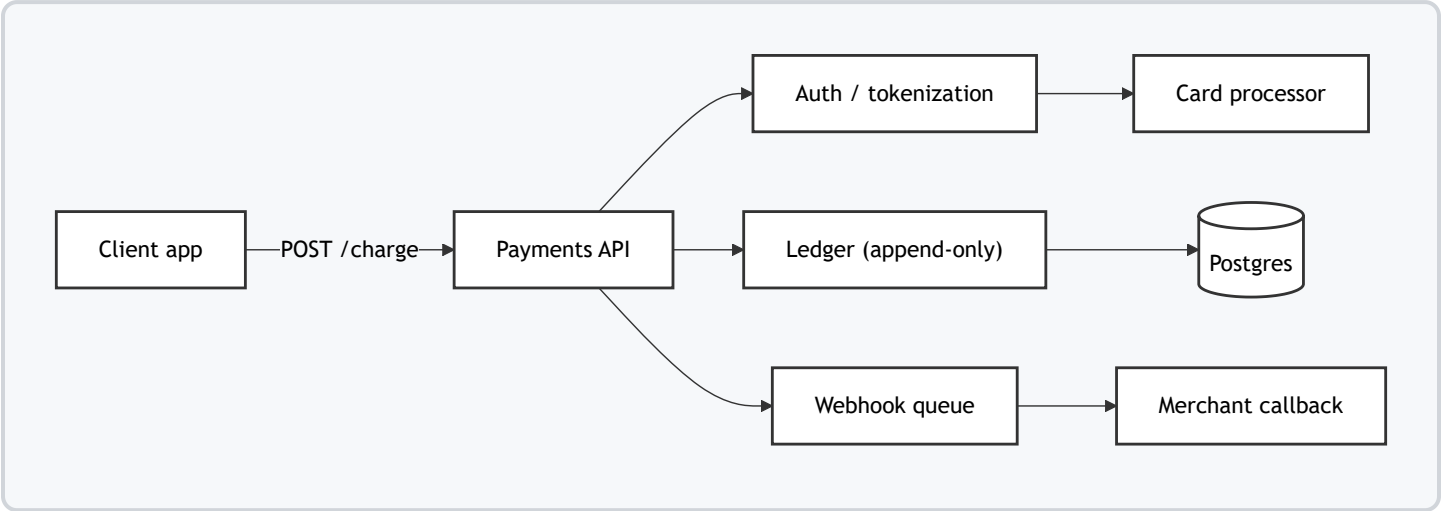
Payments Service — Design Spec

A small, auditable service that authorises card payments and records every state transition.

Note

This document is illustrative. Amounts and identifiers are examples, not production values.

Architecture



Request flow

- 1. The client posts a charge with an idempotency key.
- 2. The API tokenizes the card and calls the processor.
- 3. Every transition is appended to the **ledger** — nothing is ever mutated in place.
- 4. A webhook notifies the merchant asynchronously.

```
{
  "idempotency_key": "chg_9f2a...",
  "amount": 4200,
  "currency": "AUD",
  "capture": true
}
```

State machine

State	Next	Trigger
created	authorized	processor approves
authorized	captured	capture requested
authorized	voided	timeout / cancel
captured	refunded	refund issued

Warning

Idempotency keys expire after 24h. Re-using an expired key starts a **new** charge.